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Least squares when both variables have uncertainties

Jay Orear

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of the assembly is attached to the speaker cone with three large drops of epoxy glue. A small speaker designed for fairly large cone excursions will work best. Many small woofers are on the market for less than $10 each. The authors will furnish suppliers' names on request.

APPARATUS REQUIRED

(1) 4- or 5-in. loudspeaker with good low-frequency response, modified as described in the text. A 4-in. woofer is a good choice.

(2) A 600-Ω output impedance audio signal generator. (Less than 1 W output is required.)

(3) Transformers: (a) matching transformer (600 to 4 Ω) for matching signal generator to speaker (T_3); (b) variable transformer to control the level of the 60-Hz signal available to drive the stepdown transformer (T_1); (c) 115 to 6.3 V ac stepdown transformer to decouple speaker circuit from the ac line (any low-cost filament transformer will work) (T_2).

(4) Length of high-strength nylon fishing line. (We used multistrand nylon which had a linear density of 7.4×10^{-4} kg/m. A 2-m vibrating length requires a suspended mass of about 3.5 kg.)

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Least squares when both variables have uncertainties

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The standard least-squares method should not be applied to those rather common situations where both dependent and independent variables have errors. Instead, the effective variance method should be used. This method and the standard least-squares method are presented, along with a derivation of the general form of the effective variance method.

I. INTRODUCTION

Experiments in physics designed to determine parameters in the functional relationship between quantities x and y involve a series of measurements of x and the corresponding y. In many cases not only are there measurement errors δy_j for each y_j, but also measurement errors δx_j for each x_j. Many physicists merely apply the standard least-squares method, which implicitly assumes all the $\delta x_j = 0$. Such a procedure loses accuracy in the determination of the unknown parameters contained in the function $y = f(x)$, and it gives estimates of errors that are smaller than the true errors.

The above warning has been given recently in the Journal[1,2] but without providing an adequate solution to the problem. (Reference 1 treats only the special case of a straight-line fit where all the δy_j are equal and each of the $\delta x_j = \beta\,\delta y$.) The letter of Pasachoff[2] does give a reference to an earlier paper in the Journal, but this reference[3] treats only the special case of polynomial functions. The procedure described in Ref. 3 is a simple modification of the standard least-squares method, which is called the effective variance method. This procedure, so important to physicists, is not given in the standard statistics sources used by physicists.[4–9] It was pointed out in two letters[10,11] to the Journal that the effective variance method can be applied to more general functions than straight lines, but no proofs of the effective variance method have ever appeared in the Journal.

This paper not only presents a description and proof of the effective variance method, but also gives a review of the standard least-squares method as applied to functions more general than straight lines or polynomials, where the experimental points (x_j, y_j) can have differing errors δy_j. Then in Sec. III and IV each point is also allowed to have an error δx_j. The formalisms presented here are applied to a specific numerical example in the Appendix.

II. STANDARD LEAST-SQUARES METHOD

Let the function $y = f(x)$ contain m parameters α_j to be determined, where $j = 1$ to m. We make n sets of observa-

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tions (x_i, y_i), where $i = 1$ to n (n must be greater than m). The y_i are Gaussian distributed with standard deviations δy_i. Here the δx_i are all assumed to be zero. Then the α_j can be determined by minimizing the weighted sum of the squared deviations:

$$S = \sum_{i=1}^{n} \left(\frac{y_i - f(x_i)}{\delta y_i} \right)^2. \tag{1}$$

It can be shown that the sum S (when minimized) is the chi-squared distributed with $(n - m)$ degrees of freedom[4–9] having a mean $\overline{S} = (n - m)$ and a standard deviation $[(S - \overline{S})^2]^{1/2} = [2(n - m)]^{1/2}$. These relationships are useful in testing for goodness of fit.

Standard numerical methods may be used to locate the minimum of S in m-dimensional space. In many cases the function $f(x)$ can be put into a form that is linear in the unknown, i.e.,

$$y(x) = \sum_{a=1}^{m} \alpha_a f_a(x).$$

Then there is a simple analytical expression for those values of α_j that minimize S. If α_i^* are the least-squares solutions, then

$$\alpha_i^* = \sum_{a=1}^{m} \sum_{b=1}^{n} \frac{y_b f_a(x_b)}{(\delta y_b)^2} (H^{-1})_{ai}, \tag{2}$$

$$\overline{(\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*)} = (H^{-1})_{ij}, \tag{3}$$

where the matrix H_{ij} is

$$H_{ij} = \sum_{a=1}^{n} \frac{f_i(x_a) f_j(x_a)}{(\delta y_a)^2}. \tag{4}$$

We see from Eq. (3) that the errors in the determination of the parameters are elements of the inverse of the matrix H. In those cases where the α dependence is more complicated, a computer can be used to minimize S by iteration. Then the errors in the α_i can be obtained using the relation

$$\overline{(\alpha_i - \alpha_i^*)(\alpha_j - \alpha_j^*)} = \frac{1}{2} \frac{\partial^2 S}{\partial \alpha_i \partial \alpha_j}.$$

III. EFFECTIVE VARIANCE METHOD

The above standard least-squares method should be used only when $\delta x_j \ll \delta y_j$ for each j. Otherwise, as shown in Sec. IV, one must replace the weighting factors $(\delta y_j)^{-2}$ in Eq. (1) with $(\delta_j)^{-2}$ where

$$\delta_j^2 \equiv \left(\frac{\partial f}{\partial x} \right)_j^2 (\delta x_j)^2 + (\delta y_j)^2. \tag{5}$$

Equation (1) then becomes

$$S = \sum_{j=1}^{n} \left(\frac{y_j - f(x_j)}{\delta_j} \right)^2. \tag{6}$$

We see that the standard least-squares computer programs may still be used. In the case where $y = \alpha_1 + \alpha_2 x$, one may use what are called linear regression programs; where y is a polynomial in x, one may use multiple polynomial regression programs. The usual procedure is to guess starting values for $\partial f / \partial x$ and then to solve for the parameters α_j^* using Eqs. (2) and (4) with δy_j replaced by δ_j. Then new $(\partial f / \partial x)_j$ can be evaluated and the procedure repeated. Usually only two iterations are necessary.[3,10] The effective variance method is exact if $\partial f / \partial x$ is constant over each region δx_j. This means it is always exact for linear regressions. The numerical example in the Appendix is first solved using the standard least-squares method, then the effective variance solution is obtained and compared with the exact solution.

IV. PROOF OF THE EFFECTIVE VARIANCE METHOD

We shall use the maximum likelihood method to derive the effective variance method. In the maximum likelihood method one first calculates the joint probability of getting a particular experimental result (a set of x_j, y_j) given the true values of the α_i. Once this expression or likelihood function $\mathscr{L}(\alpha_i; x_j, y_j)$ is constructed, the α_i are then treated as independent variables. The solutions are those α_i which maximize $\mathscr{L}(\alpha)$. It has been proven that no better estimators of the α_i can be found.[12]

In our case, for each true value x_j^0 there will be a measured δx_j Gaussian-distributed about x_j^0 with a standard deviation δx_j, and similarly a measured y_j Gaussian-distributed about $y_j^0 = f(x_j^0)$ with standard deviation δy_j. Thus the likelihood function is

$$\mathscr{L} = (\alpha) = \prod_{j=1}^{n} \frac{1}{\sqrt{2\pi}\delta x_j} \exp\left(- \frac{(x_j - x_j^0)^2}{2(\delta x_j)^2} \right) \frac{1}{\sqrt{2\pi}\delta y_j} \times \exp\left[- \frac{1}{2} \left(\frac{y_j - f(x_j^0)}{\delta y_j} \right)^2 \right].$$

The logarithm is

$$ln \mathscr{L}(\alpha) = -\tfrac{1}{2} S(\alpha) + \text{const},$$

where

$$S(\alpha) = \sum_{j=1}^{n} \left[\left(\frac{x_j - x_j^0}{\delta y_j} \right)^2 + \left(\frac{y_j - f(x_j^0)}{\delta y_j} \right)^2 \right]. \tag{7}$$

In addition to the m values of α_i, there are n values of x_j^0 or a total of $(m + n)$ unknowns. The total number of experimental points is the n values of x_j plus the n values of y_j. We see that the number of degrees of freedom are $[2n - (m + n)] = (n - m)$, which is the same as before. Each x_j^0 is obtained from the single equation $\partial S / \partial x_j^0 = 0$. (The n variables x_j^0 are separated.) These x_j^0 are then inserted into Eq. (7) and solutions for the α_i are found by minimizing $S(\alpha)$. The exact solution is found by iteration. This procedure works for any functional form of $f(x)$.

Now we shall arrive at the effective variance method by making the approximation

$$f(x_j) = f(x_j^0) + (x_j - x_j^0) f_j' \quad \text{where} \quad f_j' \equiv \left(\frac{\partial f}{\partial x} \right)_{x = x_j^0}. \tag{8}$$

The condition $\dfrac{\partial S}{\partial x_j^0} = 0$ gives

$$- \frac{2(x_j - x_j^0)}{(\delta x_j)^2} - \frac{2\{y_j - [f(x_j) - (x_j - x_j^0) f_j']\} f_j'}{(\delta y_j)^2} = 0.$$

On solving for x_j^0, we obtain

$$x_j^0 = x_j - \frac{(\delta x_j)^2 [f(x_j) - y_j] f_j'}{\delta_j^2}, \tag{9}$$

where, as before,

$$\delta_j^2 = (f_j')^2 (\delta x_j)^2 + (\delta y_j)^2.$$

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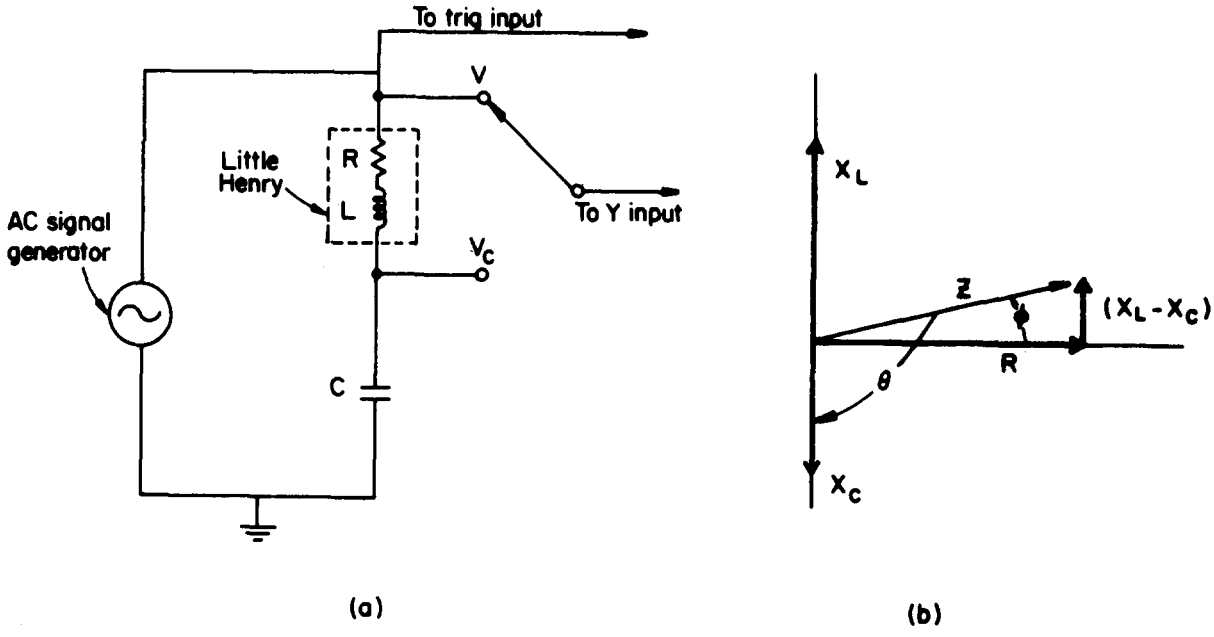

Fig. 1. (a) Schematic diagram for circuit used to measure phase shifts of V_c with respect to V. (b) Phasor diagram showing the phase angle θ measured in (a).

Substitution of Eqs. (8) and (9) into Eq. (7) gives

$$S(\alpha) = \sum_{j=1}^{n} \left(\frac{\{(\delta x_j/\delta_j)^2 f_j' [f(x_j) - y_j]\}^2}{(\delta x_j)^2} + \frac{(y_j - \{f(x_j) - (f_j' \delta x_j/\delta_j)^2 [f(x_j) - y_i]\})^2}{(\delta y_j)^2} \right)$$

$$= \sum_{j=1}^{n} \left(\frac{y_j - f(x_j)}{\delta_j} \right)^2,$$

which is the same as Eq. (6).

This is just the standard least-squares sum with m parameters α_i, and n measurements y_j, but with an effective variance δ_j^2 that is larger than $(\delta y_j)^2$. The α_i are found by minimizing $S(\alpha)$ while treating the δ_j as constants and iterating. The errors in α_i are easily obtained from Eq. (3) by replacing the δy_a in Eq. (4) with δ_a. Any standard least-squares program should still succeed provided it is iterated one or two times.

ACKNOWLEDGMENT

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APPENDIX

As an illustrative example I have chosen an experiment (with some modification) from Physics 213 at Cornell University.[13] The goal is to determine the inductance L and the internal resistance R of a black box (called Little Henry) by connecting a known capacitor C in series and applying a sinusoidal voltage V at several frequency settings. The circuit and its phasor diagram are shown in Fig. 1. First the applied voltage V and then the capacitor voltage V_c is connected to the y input of an oscilloscope whose sweep is triggered by V. The phase shift θ of V_c with respect to V is easily read off the scope face. In a series ac circuit we have

$$\tan \phi = (X_L - X_c)/R,$$

where

$$\phi = -(\theta + \pi/2).$$

So

$$\cot \theta = (X_L - X_c)/R = (L/R)\omega - (1/RC)\omega^{-1}, \quad (10)$$

where θ and ω are the measured quantities. $C = 0.02\,\mu$F is known, and L and R are the unknowns. Now let $y \equiv \cot\theta$, $\alpha_1 \equiv \omega_0 L/R$, $\alpha_2 \equiv 1/(\omega_0 RC)$, and $x \equiv \omega/\omega_0$, where $\omega_0 \equiv 1$ rad/s. Equation (10) becomes

$$y = \alpha_1 x - \alpha_2 x^{-1}. \quad (11)$$

The standard deviations δy_j and δx_j were obtained from repeated measurements of the quantities in question. Actual measurements made at five different frequencies are listed in Table I and plotted in Fig. 2.

The dashed curve in Fig. 2 is the standard least-squares solution where the δx_j are set equal to zero (labeled y on x). For comparison a standard least squares can be performed where y is considered as the independent variable and the δy_j are set to zero (x on y curve).[14] The solid curve is the solution using the effective variance method. The standard least squares takes about 20 min using a pocket calculator and goes as follows.

From Eq. (11) we see that $f_1(x) = x$ and $f_2(x) = -1/x$.

Table I. Five data points (x,y) with their rms errors $(\delta x, \delta y)$. The measured phase shift is θ_j and the corresponding frequency is $\nu_j \cdot \omega_0 = 1$ rad/s.

j	$y_j = \cot\theta_j$	$x_j = 2\pi\nu_j/\omega_0$	δy_j	δx_j
1	−4.017	22 000	0.50	440
2	−2.742	22 930	0.25	470
3	−1.1478	23 880	0.08	500
4	1.491	25 130	0.09	530
5	6.873	26 390	1.90	540

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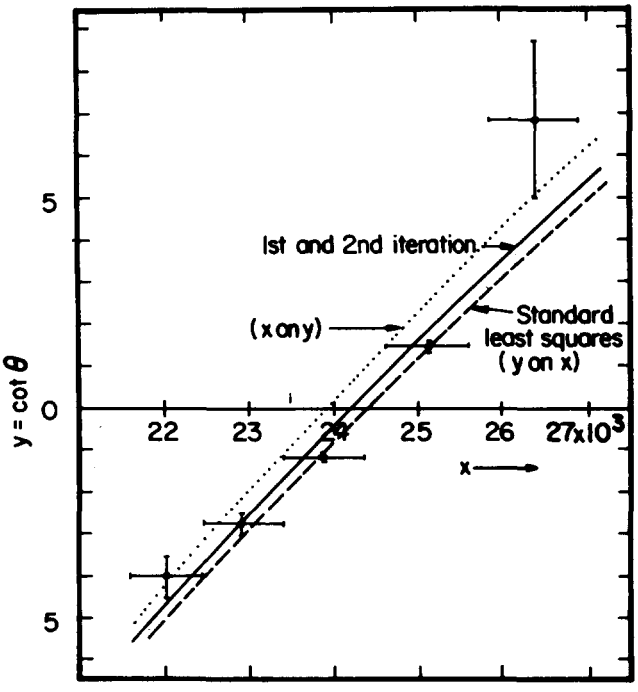

Fig. 2. Plot of y vs x showing the five experimental points with their errors in y and x. The dashed curve is the standard least-squares plot, y on x, assuming all $\delta x_j = 0$. The dotted curve is the standard least-squares plot, x on y, assuming all $\delta y_j = 0$. All iterations of the effective variance method and the exact method lie within the linewidth of the continuous curve.

According to Eq. (3) the matrix H is

$$H_{11} = \sum_{j=1}^{5} \left(\frac{x_j}{\delta y_j}\right)^2 = 1.776\times 10^{11},$$

$$H_{12} = H_{21} = \sum_{j=1}^{5} \left(\frac{1}{\delta y_j}\right)^2 = -299.98,$$

$$H_{22} = \sum_{j=1}^{5} \left(\frac{1}{x_j \delta y_j}\right)^2 = 5.0859\times 10^{-7}.$$

The error matrix is $(H^{-1})_{11} = 1.50\times 10^{-9}$, $(H^{-1})_{12} = 0.885$, $(H^{-1})_{22} = 5.239\times 10^{8}$. According to Eq. (2) the solutions are

$$\alpha_1^* = \sum_{j=1}^{5} \left(\frac{y_j x_j}{\delta y_j{}^2}\right)(H^{-1})_{11} + \sum_{j=1}^{5} \left(-\frac{y_j/x_j}{\delta y_j{}^2}\right)(H^{-1})_{21} = 9.899\times 10^{-4},$$

$$\alpha_2^* = \sum_{j=1}^{5} \left(\frac{y_j x_j}{\delta y_j{}^2}\right)(H^{-1})_{12} + \sum_{j=1}^{5} \left(-\frac{y_j/x_j}{\delta y_j{}^2}\right)(H^{-1})_{22} = 5.893\times 10^{5}.$$

Solving for R and L in terms of α_1 and α_2, we obtain $R = 1/\omega_0 C\alpha_2 = 84.84\ \Omega$ and $L = R\alpha_1/\omega_0 = 83.99$ mH. Since L is a function of both α's, the propagation of errors formula must be used for the error in L:

$$\sigma_L^2 = \sum_{a,b} \left(\frac{\partial L}{\partial \alpha_a}\right)\left(\frac{\partial L}{\partial \alpha_b}\right) \overline{\sigma_{\alpha_a}\sigma_{\alpha_b}}$$

$$= \left(\frac{1}{C\alpha_2}\right)^2 (H^{-1})_{11} + 2\left(\frac{1}{C\alpha_2}\right)\left(-\frac{\alpha_1}{C\alpha_2{}^2}\right)(H^{-1})_{12} + \left(-\frac{\alpha_1}{C\alpha_2{}^2}\right)(H^{-1})_{22},$$

$$\sigma_L = 4.63 \text{ mH}.$$

Since $\partial R/\partial\alpha_1 = 0$, the error in R is simply

$$\sigma_R^2 = \left(\frac{\partial R}{\partial \alpha_2}\right)(H^{-1})_{22},$$

$$\sigma_R = 3.295\ \Omega.$$

So the standard least-squares method gives $L = (83.99 \pm 4.63)$ mH and $R = (84.84 \pm 3.30)\ \Omega$. The least-squares sum is 10.02, which has a chi-squared probability of 0.018 for three degrees of freedom. Such a low probability is a warning that we have underestimated the errors.

In order to incorporate the errors in x, we shall now move on to the effective variance method by replacing δy_j with δ_j where

$$\delta_j^2 = \left(\frac{\partial y}{\partial x}\right)_j^2 \delta x_j^2 + \delta y_j^2 = \left(\alpha_1 + \frac{\alpha_2}{x_j^2}\right)^2 \delta x_j^2 + \delta y_j^2. \qquad (12)$$

By inserting the previous values for α_1^* and α_2^* into Eq. (12) we obtain $\delta_1 = 1.09$, $\delta_2 = 1.01$, $\delta_3 = 0.97$, $\delta_4 = 0.97$, and

Table II. Solutions for α_1, α_2, L, and R using standard least-squares, effective variance, and exact methods. S is the least-squares sum.

	α_1^*	α_2^*	L (mH)	R (Ω)	S
Standard least squares					
y on x (δx set to zero)	0.9899×10^{-3}	5.8932×10^{5}	83.99	84.84	10.02
x on y (δy set to zero)	1.0533	6.0274	87.37	82.95	6.139
Effective variance					
1st iteration	1.0095	5.9056	85.47	84.66	2.279
2nd iteration	1.0111	5.9136	85.49	84.55	2.257
convergence limit	1.0113	5.9149	85.49	84.53	2.254
Exact method					
1st iteration	1.0122	5.9203	85.49	84.45	2.251
2nd iteration (and convergence limit)	1.0141	5.9315	85.48	84.30	2.222

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$\delta_5 = 2.10$. Now we use these values for the δy in Eq. (4) and repeat everything we did before to obtain the results $\alpha_1^* = 1.01267 \times 10^{-3}$, $\alpha_2^* = 5.9222 \times 10^5$, $L = 85.50 \pm 22.48$ mH, $R = 84.43 \pm 15.37\ \Omega$, and a least squares sum of 2.328 which now has a quite reasonable Chi-squared probability of 0.51 for three degrees of freedom. This solution is plotted as the continuous line in Fig. 2. A second iteration can be performed by using these new values of α^* in Eq. (12) to get new values of δ_j which are then used in Eq. (4). Results of the first two iterations are summarized in Table II. When plotted in Fig. 2 the results of the second iteration are indistinguishable from those of the first iteration. At least in the case of this example, convergence is rapid and the first iteration was sufficient.

As stated in Sec. IV, the effective variance method is an approximation which assumes $f(x)$ is linear over each region of δx_j. As can be seen from Fig. 2, this approximation is quite valid in the case of this example. However, just to see how it goes, we shall now follow the procedure described in Sec. IV in order to obtain the exact solution. First we must find solutions for x_j^0 from each of the equations $\partial S / \partial x_j^0 = 0$. In our case this is

$$\frac{\partial S}{\partial x_j^0} = -\frac{x_j - x_j^0}{\delta x_j^2} + \frac{\alpha_1 x_j^0 - \alpha_2 / x_j^0 - y_j}{\delta y_j^2} \times \left(\alpha_1 + \frac{\alpha_2}{x_j^{02}}\right) = 0. \tag{13}$$

We start with the α_1^* and α_2^* obtained from the effective variance method. Since each of the five equations is fourth order in its unknown x_j^0, we solve them one by one numerically to obtain $x_1^0 = 22.22$, $x_2^0 = 22.87$, $x_3^0 = 23.625$, $x_4^0 = 24.935$, and $x_5^0 = 26.68$ kHz. These values of x_j^0 are then put into the equation

$$S'(\alpha) = \sum_{j=1}^{5} \left(\frac{y_j - \alpha_1 x_j^0 + \alpha_2 / x_j^0}{\delta y_j}\right)^2$$

and S' is minimized with respect to α_1 and α_2, i.e., the standard least squares is used where the x_j^0 are used in place of the measured x_j. The results are $\alpha_1^* = 0.001\,012\,2$ and $\alpha_2^* = 592\,028$. These new values of α can be inserted into Eq. (13) and the process repeated to give $\alpha_1^* = 0.001\,014\,1$ and $\alpha_2^* = 593\,149$ with the least-squares sum [Eq. (7)] $S = 2.222$. As we have anticipated, this is so close to the effective variance method that a plot of this exact solution falls within the width of the heavy curve in Fig. 2.

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[13]Physics 213 is a second semester introductory physics course for science and engineering majors. In this experiment the students are not required to make a quantitative error analysis, but they do obtain L and R from measurements of the phase shifts.

[14]In this case

$$x(y) = \frac{y + \sqrt{y^2 + 4\alpha_1\alpha_2}}{2\alpha_1}$$

is not linear in the α's and the minimum of

$$S(\alpha) = \sum_{j=1}^{n} \left(\frac{x_j - x(y_j)}{\delta x_j}\right)^2$$

must be found numerically.

PROBLEM

A projectile is launched with velocity v_i from a point a horizontal distance D from a vertical wall. Without using calculus, find the maximum distance h of the impact point above the launch point.

Find the launch angle and time of flight. (Solution is on page 943.)

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